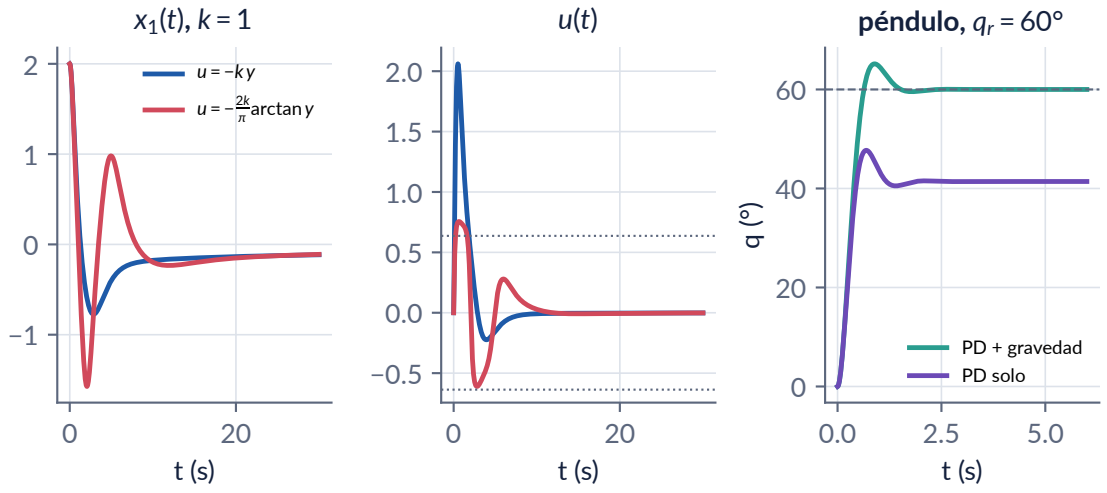
$x_1(t)$, $k = 1$
$u = -ky$
$u = -\frac{2k}{\pi}\arctan y$
2
1
0
−1
0
20
t (s)
$u(t)$
2.0
1.5
1.0
0.5
0.0
−0.5
0
20
t (s)
péndulo, $q_r = 60°$
60
40
20
0
q (°)
PD + gravedad
PD solo
0.0
2.5
5.0
t (s)